

Gym membership survival analysis

Applied Machine learning 7.5 DT4031/DT4033

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Introduction:

Business Context

The data originates from a boutique CrossFit gym in Halmstad, Sweden. Boutique gyms differ from large commercial chains through smaller membership bases (typically <200 members), premium pricing, specialized programming, and higher coach-to-member ratios. This facility maintains approximately 180 active memberships with monthly subscription fees ranging 800-1200 SEK. The membership base comprises both individual gym members and corporate clients from a contracted company wellness program.

Business Objective

The gym owner asked for data-driven insights to identify at-risk members for proactive retention interventions and quantify engagement patterns across the membership base. Given the informal operational structure typical of small businesses, the mandate was exploratory: "Analyze what you can find and report actionable insights." Key questions included identifying churn predictors, calculating customer lifetime value benchmarks, and understanding behavioral patterns (e.g., class attendance frequency) that differentiate retained vs churned members.

Project goal

Develop a predictive model for membership churn risk and identify key retention drivers, enabling targeted intervention strategies based on member characteristics and engagement behavior.

As well as finding other key business or engagement metrics.

Following data preparation and exploratory analysis (~40 hours), the project scope refined toward survival analysis as the most appropriate methodological framework, given the presence of right-censored observations (active members whose eventual churn timing is unknown) and the time-to-event nature of membership retention.

Survival analysis

Standard classification models would have treated customer churn as a binary outcome (churned or not) while ignoring the critical time aspect. Survival analysis is a branch of statistics for analyzing the expected duration of time until one event occurs, such as death in biological organisms and failure in mechanical systems. Most importantly, the framework handles what is called “censored data”. In this particular case censoring can be encapsulated in the following phrase “We know the customer is a member today, *but not if they will quit tomorrow*”.

This framework is ideal for analyzing customer churn and risk of quitting.

Data description

Data was pulled from the gyms administrative system (fig 1. brpsystems.com). Meaning the data was not created with the intent of measuring customer retainment metrics.

The SaaS system had been operational since 2021 based on looking at training class dates, however the gym was established 2012. It was unclear what quality and amount of data was available from pre migration of the current SaaS system at the time of downloading.

A total of 10 .csv files were downloaded, cleaned and merged, see appendix for details on methods, code is omitted due to privacy concerns.

There were data quality issues such as weird entries, customers with no payment history that should have (most often another family member paid double in those cases), staff, members being promoted to staff mid tenure, changed backend system functionality and measurement capabilities.

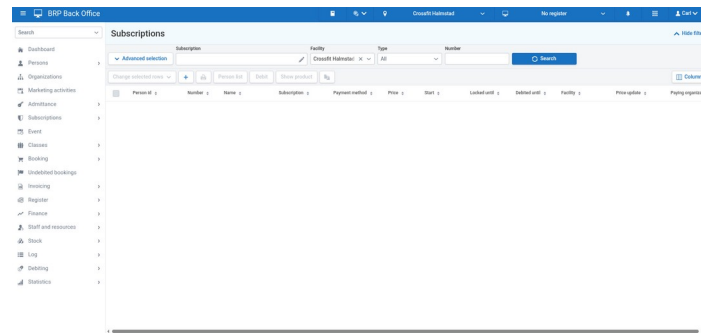


Figure 1: Censored view of the backend system

Close communication with owner and staff was held regarding data cleaning.

Anonymization code was made using the uuid4 system in order to replace name with encoded strings (annon_id column).

Due to the data cleaning required, some real members were probably lost (see data quality issues above)

The initial shape of the dataframe was 304 observations, 22 columns and 199 inactive and 105 active members.

Columns table

Colname	dtype	Orig csv source	Description	Cleaning note
anon_id	str	Name=Join key		uuid4
birth_y_m	datetime	df1	Birth year and month of member	Extracted from personnr during cleaning
age	int	df1	Age of member	Extracted from personnr during cleaning
most_recent_visit	datetime	df1	Last time member was at the gym	Contains year, month, date hour and minute timestamp for latest visit
membership_creation	datetime	df1	When the member was added to the system. Proxy for joining as member	Contains year, month, date of membership creation (hour and minute removed during cleaning)
tot_door_open	int	df4	Total times the system registers the member opening the front door*	Aggregated data from cleaning
most_recent_y	int	df1	See above	Extracted from most_recent_visit during cleaning
most_recent_m	int	df1	See above	Extracted from most_recent_visit during cleaning
most_recent_d	int	df1	See above	Extracted from most_recent_visit during cleaning
tot_class_cancellations	int	df3	How many times a class has been canceled*	Aggregated data from cleaning
check_in	int	df7	Total training sessions (door entries + class check-ins combined)	Aggregated data from cleaning
total_itv_inc	float	df2	Summed all subscription payments per member (SEK)	Aggregated data from cleaning
total_itv_ex	float	df2	See above excluding VAT	Aggregated data from cleaning
days_since_last_pay	int	df2	Days since last subscription payment (most commonly autogiro)	Extracted from datetime days since last pay during cleaning
sub_type	str	df2	Most common subscription of the member	dame of subscription type
subscription_duration_days	int	df2	Made from first payment date and last payment date	Extracted from membership creation during cleaning
sub_cancellation_date	datetime	df6	Date subscription was canceled	Contains year, month, date of membership cancellation (hour and minute removed during cleaning)
theoretical_inactive_date	datetime	df1+2	Estimated membership end date based on last payment + 30 day period.	
			Used to calculate survival time for churned members	
n_quits	int	df6	Number of times subscription has been canceled (indicates rejoining)	Aggregated data from cleaning
active	bool	df1	Is the member currently active?	Survival analysis feature: Active="no"
event	bool	df1	Is the member churned?	For churned: membership_creation to theoretical_inactive_date.
time	int	df1 + 2	Time as member in days.	For active: membership_creation to Feb 14 2026 (censored)

* Low data quality. Staff might have manually opened door or canceled after receiving a text from the member

** Negative values indicate payment occurred after reference date (Feb 14); may occur for members on monthly autogiro billing cycles.

Figure 2: Column names, source and brief descriptions

Cleaning

`tot_class_cancellations`, `tot_door_open`, `n_quits` had all their NaN values filled with 0 as they most likely signified members never canceling a class, manually opening the door or quitting membership only to rejoin.

Eda and feature engineering

Mean age: 35.95 Std age: 8.74

Histogram shows there is no clear visual difference in age between active and non active members (fig 3).

But age might interact in nonlinear ways with other features so the feature was kept for analysis.

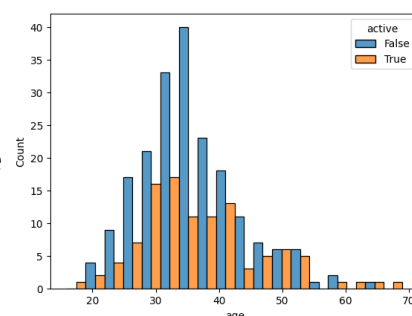


Figure 3: Age histogram color coded on membership status

There were a total of 15 types of subscriptions. Domain knowledge identified several of them as deprecated subscriptions. This led to the decision to collapse any subscription type with ≤ 9 observations into “other” category in order to decrease risk of overfitting and dimensionality issues from one hot encoding. This made 7 categories consisting of ~9% members into a single category.

Several aggregated features (`total_ltv_inc`, `tot_class_cancellations`, `check_in`) were initially considered. However, cumulative metrics conflate engagement intensity with membership duration—high values could indicate either frequent engagement or simply long tenure. To isolate engagement patterns independent of time, all counts were normalized to monthly rates by dividing by membership duration.

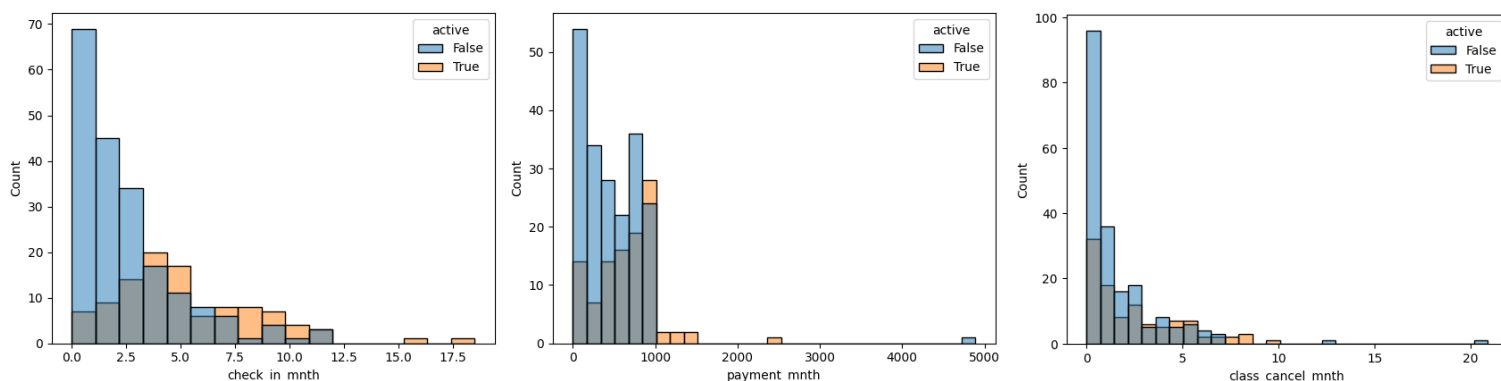


Figure 4: Normalized behavioral metrics

Plotting the histograms of the engineered features show that there seems to be differences between the distributions of active and non active members behaviors (fig 4).

Inactive members (blue) cluster at lower `check_in` frequencies (0-2.5 visits/month) and minimal class cancellations, while active members (orange) exhibit wider spread with higher visit frequencies (up to 18 visits/month). Inactive members also had more cancellations (up to 10/month). Payment intensity shows less separation but churned members display a bimodal pattern with spikes around 1000 SEK/month (monthly subscriptions) and lower rates (annual contracts or partial months). The main reason for bimodality is that prices were increased by a large margin due to negligence of annual small price increases by former owners.

These patterns support hypotheses based on domain knowledge:

High visit frequency may indicate burnout/injury risk and therefore leading to churn

Class cancellations signal disengagement

Shorter subscription durations indicate lower commitment (and also higher monthly price)

One idea for a feature to include was seeing whether there was a monthly pattern for last visit.

However, there was a high skew of last month = jan or feb since the data contains active members and they have recently visited. This made the feature unusable for modeling as it would have been a data leakage.

The other features from the table above were dropped as they were made redundant from feature engineering.

Y targets. “Time” and “Event”

The survival analysis framework requires two target variables: event (churn occurrence) and time (membership duration in days). The administrative system provided an active status field (True/False), which was used to define the event variable (event = active == 'no'). However, determining accurate membership duration proved more complex due to limited system documentation.

The system is a software service and the gym owners were unfamiliar with how the system works beyond the practical day to day operations such as booking & canceling classes and signing up new members.

This meant that no one in the gym knew how the system flagged a given member as inactive. The first attempt at finding the time variable was by calculation subscription duration days by taking each member's first and last payment date.

However, while working with attendance data it became evident that this was incorrect as there were several members who regularly attended class 2 years after subscription duration had ended. Upon further examination it became clear that once a lock in period is done, subscription switches over to monthly autogiro and membership continues without a subscription flag.

A “good enough for small business” approach was developed in consultation with the owner. Taking last payment date for a given member and add a month to that date. This will give correct time (and therefore churn) calculations for all members who had monthly subscriptions, which is the vast majority. Only a few “pay cash in full” members will have incorrect dates.

This was not enough to solve all problems. There were still 36 members missing payment date. For these members, the old way of using subscription duration was used rather than dropping ~10% of the observations.

Upon further exploration it was found that several membership creation dates were impossible and most likely placeholder values. 14 observations were dropped due to those impossible dates.

The final dataset spans 2016-2026, capturing both pre- and post- SaaS migration (2021) membership cohorts.

The final number of observations were 290.

Kaplan Meier curve

A Kaplan Meier curve was created using the scikit-survival package on all the members of the cleaned dataset (fig 5). It follows the standard -log shape and one can see the flattened part at the end before the censoring is applied.

Survival probability declines steadily through the first 2,500 days, after which the curve approaches a plateau of approximately 10–15%, indicating a subgroup of “lifer” members.

This is correct according to domain knowledge, as several members have been with the gym since its establishing in 2012.

The red markers indicate the 25th, 50th, and 75th survival percentiles and expected value for new members were calculated at those percentiles with the formula $[(\text{percentile}/365)*964*12]$, using the gym's current 12-month autogiro subscription rate.

Q1: 376 days LTV 11916 kr

Median: 972 days Expected LTV 30805 kr

Q3: 2343 days LTV 74257 kr

The flattening of the curve violates the proportional assumptions of Cox proportional hazard model, indicating that models with different assumptions will perform better. However, at least one Cox PH model will be run to acquire feature importance in order to provide business insights to the gym owner.

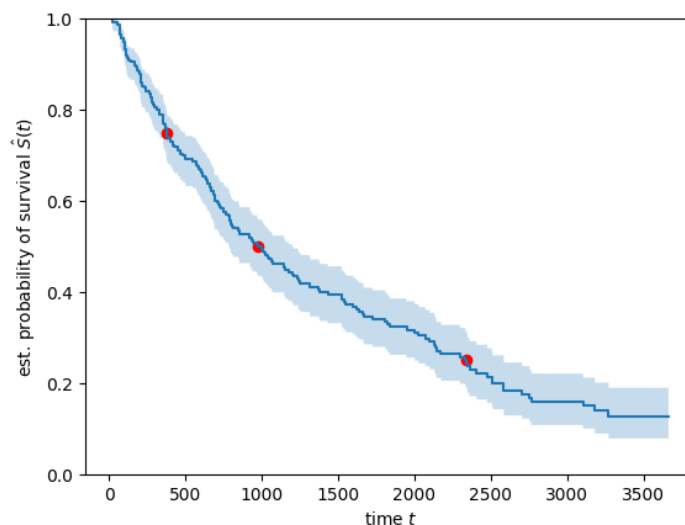


Figure 5: Kaplan Meier Curve with 1-3rd percentiles

Train test splitting and relationship with target variables

A 80% train, 20% test split was used with models trained on 5 folds of cross validation. No pure validation split was made due to the nature of the business task and relatively small dataset. A loss of statistical rigor is fine when the business owner wants general directions rather than perfect predictions.

Pairwise scatter plots

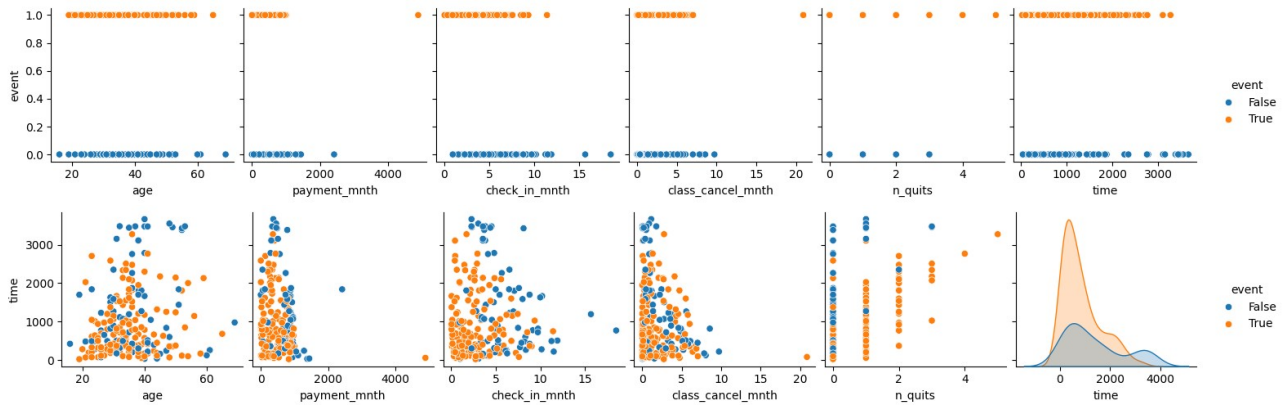


Figure 6: Pairwise scatter plots. Top: Event. Bottom: Time

Pairwise scatter plots on the training data of features against membership duration reveal substantial overlap between churned and active members across all features, with no single variable providing clean separation (fig 6). Small distributional differences are visible in payment_mnth, check_in_mnth, and n_quits, though these are insufficient for reliable univariate discrimination. This heterogeneity reflects the complex nature of gym membership churn and, as previously mentioned, indicates a need for a more complex model than Cox proportional hazards.

Linear correlation matrix

A linear correlation matrix was made (fig 7). Subscription subtypes were temporarily ordinally encoded into a single column rather than one hot encoded order to make the matrix more readable.

No problematic multicollinearity between features were found (all $< |0.5|$).

Features show weak-to-moderate correlation with time ($\sim|0.15-0.4|$), confirming that monthly normalization successfully reduced conflation with membership duration.

The strongest feature-feature correlation is check_in_mnth vs class_cancel_mnth (0.42). This is likely partially a volume effect: members who book more classes naturally accumulate more cancellations, so high cancellation counts should not be interpreted as a disengagement signal in isolation.

Event correlations are fairly weak ($\sim|0.15-0.45|$), with n_quits (0.39) and check_in_mnth (-0.5) showing strongest univariate relationships. The check in per month finding seems very reasonable is very interesting from a business perspective, members who show up fairly often don't quit. Age shows near-zero correlation with churn (-0.098), suggesting demographics are less predictive than behavioral metrics.

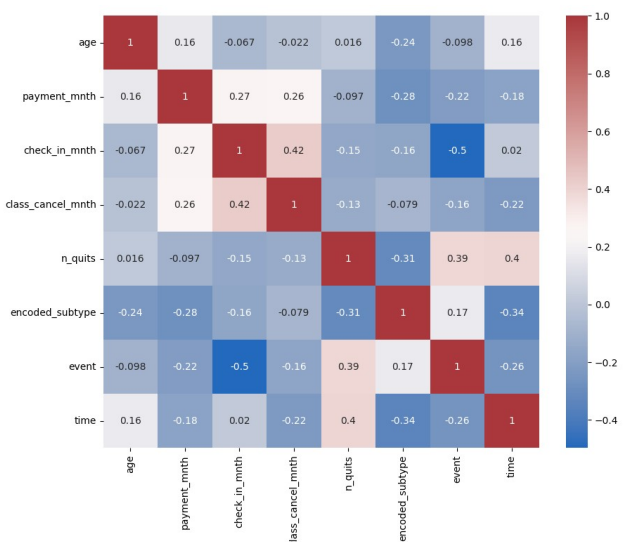


Figure 7: Linear correlation matrix

Methods

An exhaustive grid search with 5 fold cross validation of the models listed below was conducted, using concordance index as the scoring function. The small dataset (~300 observations) makes exhaustive search computationally trivial, justifying a broad parameter grid over a more targeted one. As noted earlier, the inclusion of Cox Proportional Hazards models were mostly for finding feature importance and stakeholder communication rather than the expectation of performance.

Cox proportional hazards -No tuning available
Regularized **Cox Proportional Hazards (CoxNet)**

- l1_ratio: [0.01, 0.25, 0.5, 0.75, 1.0]
- alpha_min_ratio: [0.001, 0.01, 0.1]
- Total fits of 5 cvs: $5 \times 5 \times 3 = 75$

Survival tree

- max_depth: [3, 5, 10, 20, None],
- min_samples_split: [6, 10, 15, 20],
- min_samples_leaf: [3, 5, 10, 15]
- Total fits of 5 cvs: $5 \times 5 \times 4 \times 4 = 400$

Random survival forest

- max_depth: [3, 5, 10, None]
- n_estimators: [50, 100, 150, 200]
- min_samples_split: [6, 10, 20]
- min_samples_leaf: [3, 5, 10]
- max_features: [sqrt, 5, 7]
- Total fits of 5 cvs: $5 \times 4 \times 4 \times 3 \times 3 = 2160$

Results

As expected, cox proportional hazards performed worst, both in absolute performance and when considering overfitting. Both regularized and non regularized cox.

There was a surprisingly small difference in CV scoring between the models, however random survival forest performed best on CV score, Training data and test data and marginally worse than survival tree on CV-test

	Model	CV Score	Train C-Index	Test C-index	CV-test Δ
0	Cox	0.7469	0.7688	0.7235	0.0234
1	Elasticnet Cox	0.7486	0.7687	0.7244	0.0444
2	Tree	0.7639	0.8401	0.7643	-0.0004
3	Forest	0.7794	0.8735	0.7788	0.0006

Figure 8: Model C-index comparison

No interpretable pattern was found other than trees were allowed to grow quite deep.

- `max_depth`: None
- `n_estimators`: 150
- `min_samples_split`: 10
- `min_samples_leaf`: 3
- `max_features`: `sqrt` (3)

As seen on the plots below. Cox proportional hazard predictions seemingly follow the actual training and test Kaplan Meier curves quite well, whereas random forest diverges as t increases. It is unclear at this point whether the model found some potential long term members not captured by the Kaplan Meier curve, performs worse as t increases (that statement assumes Kaplan Meier to be ground truth, which it is not) or if it is simply an unlucky artifact from a small dataset not suited for the task at hand.

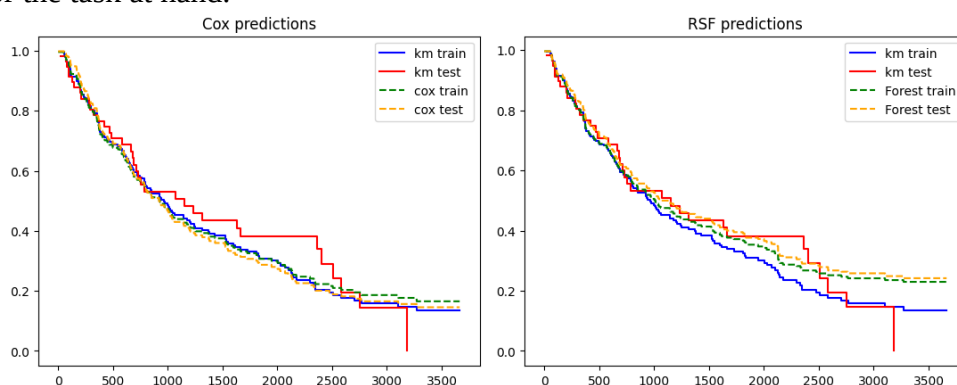


Figure 9: Cox PH & RSF predictions compared to KM curves

Cox Proportional Hazards feature importance

In line with business goals, feature importance from the Cox proportional hazard model was extracted. One can clearly see that being an “intro brons” member is by far the largest risk of quitting (fig10). This is supported by the mode subscription time found during data cleaning (see appendix for details), along with both owner and staff of the gym confirming that it is a weak point in the sales funnel. One big caveat of this observation is the fact that if an introductory member graduates into a full member, their membership status changes meaning the original membership status becomes censored. This is a major limitation to the current model, however handling nested censoring is beyond the scope of this project.

Additionally *veckokort 2* (2 weeks short membership) and *klippkort* (punch card) are also major risk factors. In retrospect these should have been collapsed into “other” category as they represent temporary membership such as summer vacation guest training from a pure modelling view, at least.

From a business perspective, however, they are semi-committed members already demonstrating engagement, and prime candidates for conversion. A simple trigger, e.g. “Prompt a membership conversation after 8 of 10 punches are used ” could capture members already demonstrating engagement before they default to buying another punch card or drifting away.

The student-type also suffer from the censoring issue as intro members. Either a student is done with studies and leaves town and the model correctly identifies an event, or they graduate into full members and the membership status gets censored.

The only real actionable insight from the coefficients is average number of check ins per month that offers a small protection from churning.

More discussion on this topic will be done in the “Future improvements” section.

Additional feature importance exploration is done in code but were omitted from this report due to the lack of actionable insights.

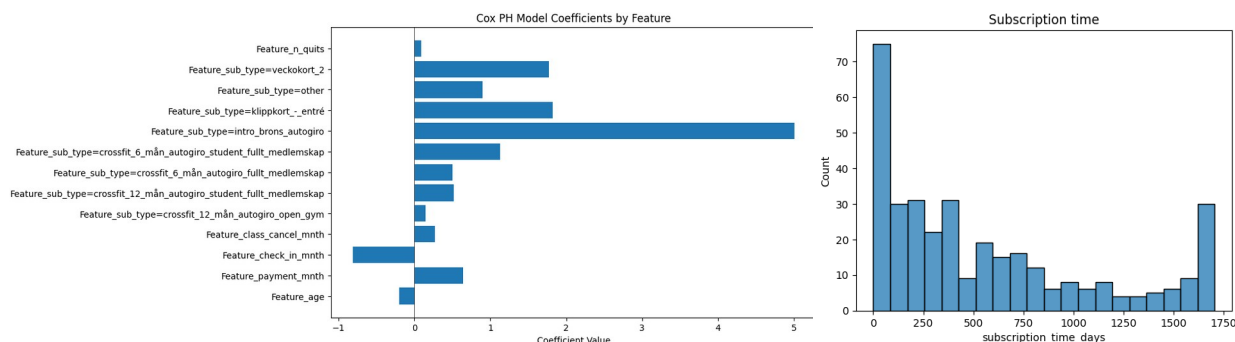


Figure 10: Cox PH feature importance & Subscription time from non-anonymized data cleaning (see appendix for details)

Limitations and future improvements

As previously mentioned, the data was not created with the intent of managing member engagement.

And the system in question has undergone changes during the lifetime of the gym.

Therefore a lot of assumptions had to be made when cleaning and merging the data prior to starting the modeling process (see appendix for details).

A big model limitation was found when visualizing top 5 churn risk members. As seen on fig 11, idx 0 and 1 observations are very early in their membership tenure.

Upon closer examination they had deflated behavioral values due to short membership periods. Therefore the model flagged their artificial behavior as risky, but their membership status indicated a lock in of at least 6 months. Plenty of time to stabilize behavior and subsequent values for the model to use.

In order to pilot the deflated behavioral score hypothesis, the visualization was remade iteratively with tenure cutoffs at >30, 60 or 90 days.

This improved predictions, see fig 12 for >90 days, but idx3 on that plot is a punch card member, reaffirming prior observations of making better membership type collapsing decisions for modeling.

Filtering members to run predictions on is at best a band aid. The underlying model has already learned from the suboptimal features. Or as the colloquial saying goes “Garbage in, garbage out”.

This project was introduced to survival analysis through direct supervision, including a dedicated lecture on the fundamentals. While this provided sufficient grounding to proceed, a more thorough engagement with the literature, such as reading Machine Learning for

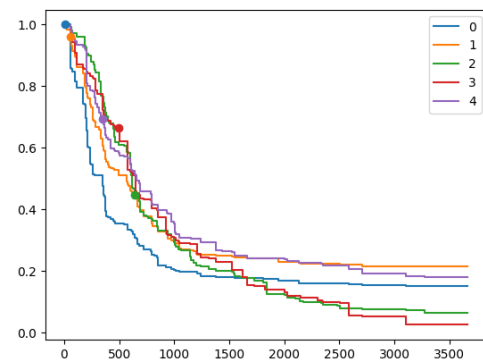


Figure 11: Initial top 5 risk survival function plots

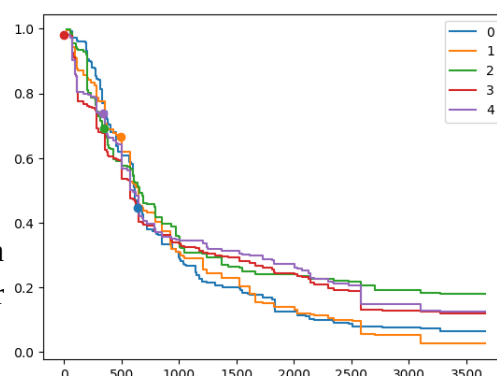


Figure 12: Top 5 risk survival function plots filtered by 90 day membership tenure

Survival Analysis prior to the data cleaning phase would have informed several design decisions that had to be addressed post-hoc.(<https://www.mlsabook.com/>)

That said. Some potential improvements would be to feature engineer behavioral deltas rather than aggregating and normalizing them. A member who has historically been engaged but has stopped attending in the past 30 days represents a qualitatively different risk profile than a consistently low-engagement member, yet both may produce similar aggregate feature values. This seems both informative from a model standpoint and actionable from a business perspective.

Conclusion

Considering the data quality issues, a model with 0.77 test C-index is quite good in aggregate and can be considered a success. However, due individualized nature of a boutique gym, individual predictions should be interpreted with caution.

The importance of improving follow up with introductory members is a clear and actionable business insight, along with the expected lifetime value numbers provided from the Kaplan Meier curve to show the importance of improved business practices. Additionally, tracking monthly check-in counts via the existing gym management system represents a low-effort, high-value operational habit.

Survival analysis is an appealing subcategory of machine learning with several interesting use cases such as predictive maintenance and drug/intervention effects on diseases and pain management. This along with the sparsity of material on the subject compared to more mainstream machine learning topics such as regression, classification and deep learning suggest a worthwhile niche to pursue and explore further.

Additionally the freedom to attain real world data provided much needed motivation as it allowed plenty of practice opportunities of all steps from the data science life cycle.

risk_score	age	time	sub_type	days_since_last_visit	class_cancel_mnth	payment_mnth	check_in_mnth
129.234971	27	653	crossfit_12_mån_autogiro_fullt_medlemskap	9.0	2.756508	782.067179	2.021440
122.032198	40	2372	klippkort_-_entré	3.0	0.012648	35.096909	0.101180
121.373779	27	578	crossfit_6_mån_autogiro_fullt_medlemskap	494.0	0.051903	646.401384	0.207612
120.231080	42	599	klippkort_-_entré	8.0	7.262104	7.512521	1.803005
111.241761	48	360	klippkort_-_entré	4.0	0.250000	291.750000	1.583333
109.574122	31	444	crossfit_6_mån_autogiro_fullt_medlemskap	3.0	2.432432	527.027027	2.770270
105.528017	43	394	crossfit_12_mån_autogiro_fullt_medlemskap	3.0	0.761421	972.335025	1.446701
97.175012	33	495	crossfit_6_mån_autogiro_fullt_medlemskap	2.0	5.090909	663.575758	6.787879
91.220166	16	422	klippkort_-_entré	3.0	3.696682	63.981043	5.260664
89.154665	29	816	crossfit_6_mån_autogiro_fullt_medlemskap	2.0	4.889706	767.022059	5.110294

Figure 13: Final, filtered, risk members along with behavioral features

Appendix

Cleaning and merging process notes. Code omitted due to privacy concerns.

Data cleaning notes round 1

DF1

df= persons_all_columns.csv

<https://crossfithalmstad.brpsystems.com/backoffice/person>

Should contain data of both active and “inactive persons”

Unclear what inactive means

Mostly demographic (sensitive) data.

Several columns are more or less empty.

Customer no can possibly be used for anonymization. But I think full name has to be maintained in order to merge dfs.

Following columns were removed as they contained no or sensitive information:

```
df=df.drop(columns=["Organization", "Reference code", "Category", "Price list", "Card number", "First name", "Last name", "E-mail", "Phone", "Address", "Zip code", "Groups", "Local groups", "Customer type", "Customer type ends", "Sales status", "Mobile", "Personal number"])
```

“Birth date” was converted to “age” and “birth_y_m” and subsequently dropped in order to increase anonymization while preserving valuable customer information.

Facility column contained information regarding which facility of the gym the member in question was training. After talking to stakeholders it was decided to remove anything related to “Hallands Hamnar” as that was a temporary contract with that company.

Additionally it was agreed to split data into “Crossfit Halmstad” and “AJ produkter”, since there were different questions regarding those customer groups/facilities. “Crossfit Halmstad” contains the data relevant to the actual gym.

One thing that stood out was the large amount of missing values in “Most recent visit”. Maybe tracking was bad?

After checking a few specific members last visits (only possible due to me knowing half the gym members) it was concluded that the tracking was accurate in those cases.

After further digging an inactive_no_visit df was built from

```
(df_cfh['Active'] == 'No') & (df_cfh['Most recent visit'].isna())
```

When investigating the names on that list I came to the conclusion that those members quit before the gym switched door opening system from a card to an app with bluetooth connection to the door, indicating better tracking capabilities from there on.

These pre app members were subsequently dropped from further analysis.

The second column with high missing values was "Town" (32.4%). Missing data showed no correlation with member status (Active vs. Inactive), and populated values showed inconsistent

formatting (varied spelling and whitespace). As "Town" is an optional field during signup often completed in haste before group classes and showed no predictive value for churn, it was dropped from the analysis.

Six members with missing age data were identified as non-Swedish residents without personnummer. These were excluded as age is a key demographic feature.

DF2

all_bookings.csv

It is a little unclear what this table contains. There seem to be pricing data (interesting!) but also whether different rooms of the gym is booked.

Several columns were dropped as they contained no information at all, was filled with the exact same value, or the values were unclear what they represented ("length" contained values between 0-4 but most were null)

"Organization", "Company", "Facility", "Number of units", 'Resources', 'Resources (staff)', 'Resources (non staff)', 'Internal message', 'External message', 'Internal notation', 'External notation', 'Units', 'Length', 'Department', 'Preliminary', 'Number of participants', 'E-mail address', 'Channel', 'Commission recipient', 'Mark', 'Dimension 1', 'Dimension 2', "Booking labels"

The columns "Responsible" and "Orderer" contained conflicting information. At face value "Responsible" would be the person in the checkout receiving the order and/or payment. And "Orderer" would be the customer buying.

But checking values in "Responsible" revealed that there are just as many members as staff "Responsible" perhaps this is a matter of auto renewal of membership? Have to check with gym owners.

Same confusions with "Orderer" its an even split between staff and members here as well. But crucially. Not the same as in responsible column.

This whole DF might be a mix of all kinds of orders. Such as booking rooms for group classes by staff, refilling membership and buying energy drinks from the fridge.

Looking at value_counts in Products confirm this:

Group class bookings (CrossFit, Hyrox, etc.) - 16,087 CrossFit bookings!

Membership payments ("12 mån Autogiro") - ~3,000 transactions

Snacks/drinks (Nocco, protein bars, Pepsi Max)

Locker rentals.

There were also several rows with no price. The first 25 rows of the most common "Product" value were class and room bookings.

```
df2[df2['Price'] == 0]['Product'].value_counts().head(25)
```

They were subsequently dropped.

Investigating further by repeating the code line above on the partially cleaned df (equivalent to entries [25:50]

and using domain knowledge showed that the remaining values in all likelihood were any of the

following:

Bookings, Staff free membership, membership autogiro initialization, freebies and gifts.

This led to simply dropping every observation where price was == 0.

The dropping also led to the removal of strange products with names such as “1.0” and “0.0” that were filled with NaN values. Probably some form of systems testing product.

All products sold 5 or fewer times were also dropped. Initial inspection suggested a cutoff of around 20, however this range contained legitimate recurring membership payments which would have been incorrectly removed. Subsequently all remaining products had to be hand picked based on membership in mutually exclusive categories. Categories selected were

- “products_to_drop” witch consisted of items sold in small quantities with low value.

Additionally deprecated products were added here as they have no bearing on current day sales.

- “products_sales_analysis” witch consisted of products frequently sold but not tied to any customer name. Additionally, the gym has a separate self service system that isn’t connected to this data. Making the data unreliable to model on.

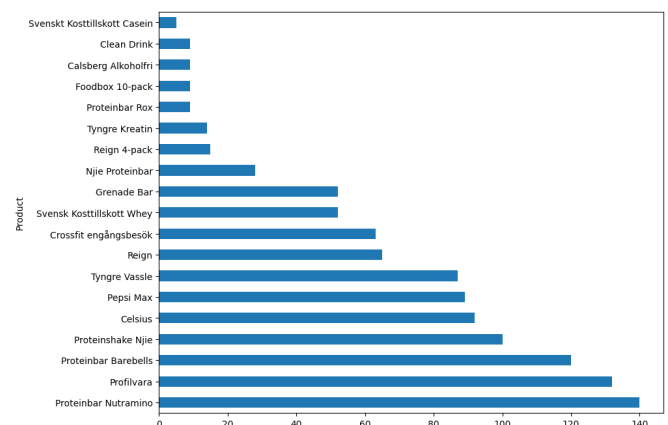
- “churn_ltv_targets” consisting of payments regarding membership tied to a member

As the product list is ~50 entries long I wont detail everything in this report.

46 cells in price incl VAT were missing. Fortunately the excl VAT price and VAT rate was available for them all. No further nans were found after that.

The only column with nans left was “Responsible”. Seemingly this column is related to staff members operating the store, but there were member names mixed in there, thus making the column information unreliable.

Column is kept as is until further information is available.



Df3

all_cancellations.csv

As the name suggests, this file contains class cancellations with names of the members canceling.

The following were dropped in immediatly:

Cancellation types' – All either nans or “giltlig”

Order' – Order no for system processing

'Resources' – Concerns wich room the class was held, irrelevant for churn

'Cancellation fee', 'Fee debited'-These only contained nans or 0 thanks to laissez-faire cancellation policy

No further nans were found.

Created and Start were both kept as they seemed to represent when the cancellation took place and when the class in question were to start.

Potentially one could build time series analysis on cancellations for a given member.

Or see what days most classes got canceled.

The top class cancellation member was one of the owners (1668 cancellations, 2nd “top” canceler had 401). Some staff members were found in the top 10 cancelers. After talking to the gym owner it became clear that this was due to manual cancellation of other members at the front desk. The scenario given was “If X text me on messenger and I cancel at the desk, the cancellation will count towards myself”

The staff values were removed leading to a reduction of 2622 from 20668 total cancellations.

Sadly this means the faithfulness of the cancellation data is limited. But cancellation data seems like an important feature for future predictive models, so it was kept despite lack of data quality.

DF4

all_entrance_logs.csv

This df concerns door openings. Reliability of this data is not very good since entrance wont be logged if someone else has opened the door.

Rows where Results != “OK” were dropped as those were either Errors, expired membership, wrong timeslot for groupclass, wrong card used. Or in the case of one former member, suspension. It is unclear whether this df will be used given the unreliability of the data, but it might be worth keeping for other metrics or trivia such as “Guess who has entered the door the most?” for member of the month type content.

The column itself was dropped after rows were deleted.

Rows where New visit = “No” were also dropped as they signified a member entering more than once on a given day.

The odds of a double session are considerably lower than incidental re-entries such as returning to a car for forgotten equipment. This assumption is based on domain knowledge of typical gym member behaviour.

Rows with “Yes” were maintained as this kept the information whether the member entered that day.

After that the column was dropped.

Rows where there was any text value in Message was deleted as they were only concerning punch card usage and therefore were irrelevant to subscribing members. After rows were deleted, the column was dropped.

When looking at “Reader” column there were 2 values. “Ytterdörr” and "Bokning". The second signifies coming to a class, and if there is any point of the day where a member might be let in by someone else its around class. So this column will make unreliable data even more unreliable. Therefore it was dropped.

Df5

All_paid_invoices

As the name suggest. Contains all paid invoices.

This is a great source of lifetime value for a member.

As well as checking when they paid their last invoice.

Organization contained only null values so it was dropped

Payment Journal seemed to contain batches of payments for a given day. It was dropped as it related to accounting rather than members.

Same with Invoice Number. Accounting infrastructure. Dropped.

Payment method contained 3 different values. Bank, Extern Kort and Nan.

Extern kort had only 4 observations and the prices were the same as Bank prices.

Nan values were double entries of same customer on the same day with a positive price value in the first and the same value but negative in the second entry. Probably repayment due to cash register mistakes.

Therefore NaN rows were dropped (repayment errors), Extern Kort and Bank rows were both retained, and then the column itself was removed since the payment method distinction was no longer needed

Df6

all_subscription_cancellations.csv

This is where the main churn data will come from

Name, Subscription (values seem to match with the product names regarding subscriptions from df2!),

Cancellation date is the big one of interest.

All good stuff!

Name column had to be sorted as it was Surname Lastname.

Sadly Reason for cancellation contained very little information. Nan and "På egen begäran" (tr: upon request) accounted for over 300 of the 360 rows

42 were "Flyttat" (moved to new city), 5 "Sjukdom" (sick leave), and 1 "missnöjd" (dissatisfied).

This column had to be dropped considering how little information can be gleaned from those values.

Df7

all_visits.csv

Check out and Entry type columns were dropped as they contained only nans.

Reader contained several nans as well. First hypothesis was that they would match with those from df1 and be due to the switching of card to bluetooth door opening system.

However, checking the latest dates disconfirmed this as they were well passed the switching time period.

The second hypothesis was that the column contained a mix of door openings and class checkins.

If that was the case one would see spikes of nan values around class times on a histogram. ~7, 12 and 16-19 swedish time. And that was indeed what the histograms showed.

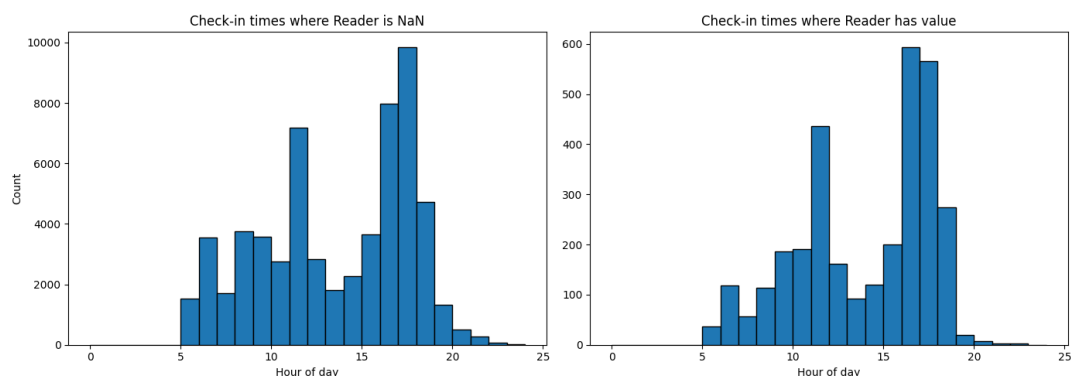
The last thing that could be suspicious would be if someone first opened the door and then checked in separately. This should lead to double entries in close temporal proximity. A non-granular check on whether a customer Checked in with a reader value and a Nan on the same day was performed. A total of 16 dates with double (or more) entries of a single member were found. As previously noted in df4, double entries most probably signifies a member running out to fetch forgotten equipment and re-opening the door. Therefore duplicate same-day entries were removed, keeping only the first check-in per member per day. After that Reader and Card number columns were dropped as they contained no more information. Along with the temporary hour and date columns.

Df8

current_members.csv

Subscription details on all active members using the Crossfit Halmstad facility

Facility column was dropped as all cells had the same value



Price update was also dropped as it only contained planned future price updates for 23 out of approximately 180 members. As these were projections rather than historical data, they provided no insight into past behavior or current metrics.

Payment method was dropped as it only contained info on whether the payment was autogiro or a single payment. And that information was already available in the values of Subscription column. "Number was dropped as it seemed to refer to accounting. Locked until was dropped as it referred to the initial contract lock-in period, which had already expired for most members and provided no information on future churn risk. Start and Debited until were retained as membership tenure and paid-through dates are relevant churn predictors.

Rows where price = 0 were dropped as those were “free” staff and/or family memberships.

df8 contains 8 members not present in the cleaned dataset, probably due to data quality filters applied earlier. There was no real pattern in those few observations so they were considered noise and dropped.

DF9

entrances_per_person.csv

Pre-aggregated entrance counts per members! No need to manually build that feature

It is a bit of overlap with earlier dfs such as most recent visit from df1. But that might be a good

thing for future joins.

Initially it seemed like the df had to be split between crossfit halmstad and AJ produkter when looking at the values in Facility(person). But there were only 8 entries from AJ. Not sure why but since it was only 8 out of 180 rows they were dropped. It is unclear why df8 and 9 (139) are of different shapes. Facility(person) was subsequently dropped as all values were the same, crossfit halmstad.

Organization column was dropped as there were only nans there

Customer no was dropped as it was not consistently present across other dataframes, making name the only viable join key.

Most recent visit, Number of visits, Name were kept

DF10

subscriptions_all_columns.csv

This df is possibly redundant as it seemingly adds no information over invoices + current members. Basically an intersection of those 2

Checking length and columns of both df8 and 10 revealed they were almost identical. Df10 was therefore excluded from further analysis

Data cleaning notes round 2: Merging

df1 all_persons

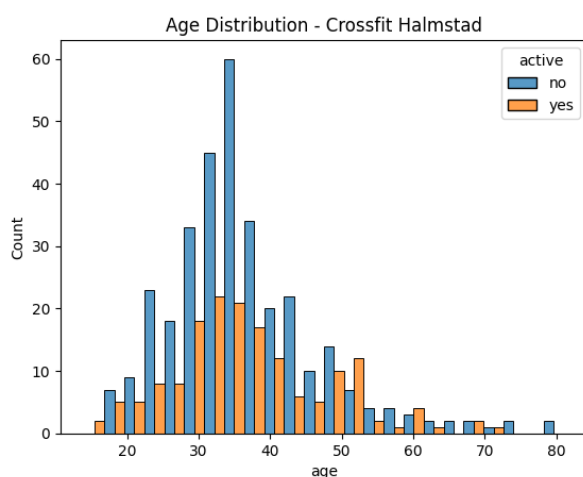
"facility", "person_id" were dropped as they were missed during first round of cleaning

Datetime conversion was made regarding most_recent_visit

Taking a look at the only real feature where statistics is relevant, age concerning active vs former members.

Most members being between ~25-45 looks very reasonable for crossfit demographics.

There are small differences but both mean and std are close to 1 year from each other, implying age alone won't be a strong feature for modeling.



```
'Current members'

count    162.000000
mean      37.092593
std       11.083425
min       14.000000
25%       30.000000
50%       35.000000
75%       43.000000
max       70.000000
Name: age, dtype: float64

'Former members'

count    324.000000
mean      36.401235
std       10.677055
min       17.000000
25%       30.000000
50%       35.000000
75%       41.000000
max       81.000000
Name: age, dtype: float64
```

Df2 bookings

Datetime conversion was made on start and end columns.

Visualizing all payments shows the majority cluster around 900-1000 SEK (standard monthly memberships) the first cluster of ~800 was an old price point, reflecting the historical nature of this data. There were several small outliers representing 'last half month' and locker rental (~50 SEK) type payments. They were kept and subsequently rolled up in the lifetime_value per member using .groupby.

The most common “product” (subscription type) was also kept.

Several members transitioned from paying customer to staff, visible as payment histories that abruptly end despite continued gym presence.

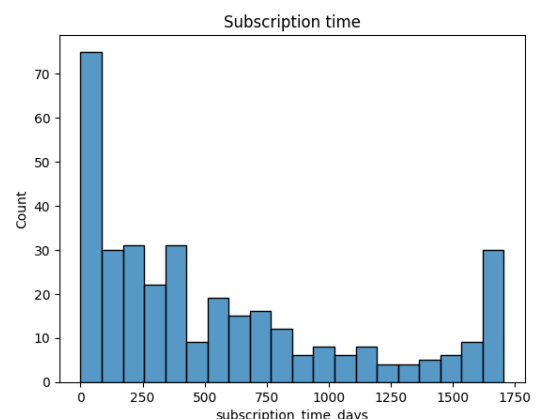
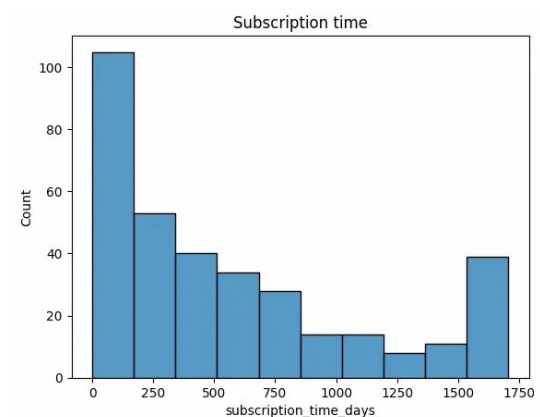
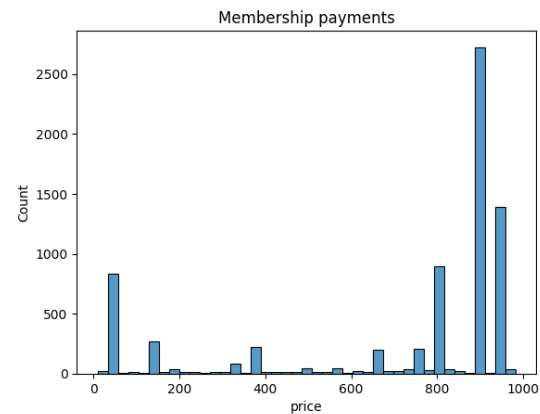
This will create apparent 'churn' in payment data for members who became staff, artificially inflating churn signals in downstream analysis.

One thing that really stood out was subscription time (consisting of start.max-start.min counted in days).

Theres a clear drop off at ~4 months. But doing a slightly more granular analysis reveal that the actual big drop off time is at around ~85 days.

From a bussiness perspective, some form of action around 60 days seem like a good idea.

A secondary peak at 1500+ days (~4+ years) represents long-term loyalists who have remained engaged past the critical early churn window.



Df3 cancellations

An aggregation of how many cancellations each customer has made was performed.

It was considered to do feature engineer some form of cancellation trend feature by calculating how many cancellations there had been in the last month compared to the second last month. However, that feature would have required cross referencing with whatever date the member quit if it was an inactive member.

This potential feature was decided against for now given the complexity of such a task. This could be revisited later if time permits.

Df4 entrance logs

An aggregation of how many entries each customer has made was performed.

This ended up with a total of 483 rows. As of writing, the current merged master df has 486.

This minor mismatch is probably due to inconsistency regarding staff removals from prior cleaning.

These members will receive NaN for entry counts and can be handled during modeling (imputed as 0 or excluded)

Df5 paid invoices

A comparison between df5 and 2 was made regarding what dates and customers were contained in each table.

There was a few days difference between first and final dates, probably an artifact of accounting and bank lead times.

There was also a big overlap in the names contained in both (294), with an additional 52 only in df2 and 16 in df5.

The marginal value from adding 16 more to the already existing 346 observations in df2 did not justify the additional effort of adding them to the final df.

DF6 cancellations

The same kind of comparison was made between this df6 and df2 as was done with df5.

There were an additional 76 members in df6 so it was worth investigating whether any features were extractable.

Digging further revealed 48 members (10%) had quit and rejoined at least once, with 15 quitting 3+ times. A feature counting total quits (n_quits) was extracted to capture this repeat churning behavior pattern. Additionally, most recent cancellation_date was retained for churned members.

Df7 visits

An aggregation of how many visits each customer has made was performed.

This might be somewhat redundant to total entries but as noted in cleaning docs. Entries and visits seem to contain different types of information (door openings and class checkins)

As a side note. This df seems like a candidate for future iterations of modeling to extract more data from the check_in column, as it contained both year, month, day and time of day data. But the marginal gains from such a project seem small, relative to get any model up and running.

Df8 current members

After investigating and comparing to the merged dataframe it was decided that this df was more or less redundant. Columns included: 'subscription', 'price', 'start', 'debited_until'

All of which are renames of features already included in merged df. The one feature that could be of value was debited_until. But 90% of the values were this month's payment (mid-late feb and early march), with just a few outliers paying larger sums in cash for longer membership stints.

Df9 entrances per person

Df9 was judged redundant as it contained pre-aggregated visit data containing most_recent_visit and number_of_visits. Both features already existed in the merged dataframe.

DF9 was subsequently dropped

Merging

df1 and 2

A left join was made on df1 as payment data was only concerned subscriptions in df2. The 178 without payment data include staff/honorary memberships, legacy members with data quality issues (placeholder dates like 1990-01-01), and likely non-recurring payment types not captured in df2. Additional LTV data from other transaction sources will be incorporated in subsequent merges.

Adding df3

Another left join was performed. No new rows were made. Only the information regarding total cancellations was added.

Adding df4

A left join was performed.

As noted in cleaning notes. The row match was very high with the merged df.

Df 6 and 7

Both were done with a left join

Final merged df

The final merged df had the shape of 486 rows and 19 columns totaling 9234 cells. 1621 of the cells were nans. The majority of them represent 'no record' rather than missing data (e.g., members with no cancellations, no payment history) and will be imputed appropriately during modeling prep 162 members were active and 324 inactive. Leading to a class imbalance of 324/162=2/1. This might change during cleaning.

Some “created” values were clearly erroneous such as before the current system was implemented.

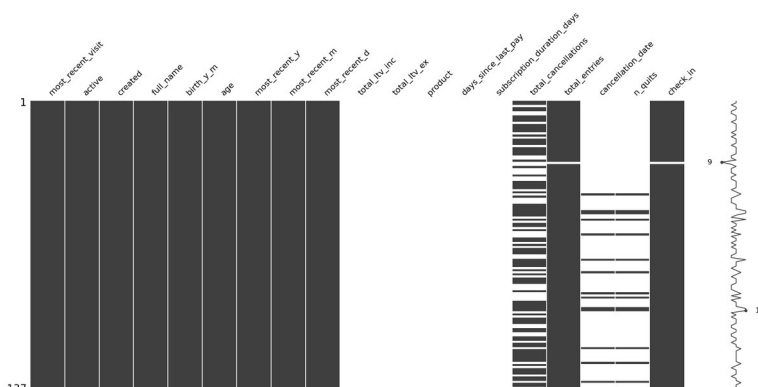
It also contained several names that were identified as staff with the firstname_lastname_(staff) handle, but also several names that were identified as current and former staff from domain knowledge. They were removed with this anonymized code:

```
staff_remove_these=no_payment[no_payment["full_name"].str.contains("staff|xxx_performance|name1|name2|name3", case=False, na=False, regex=True)]
```

There were several data quality issues found when investigating the names, especially regarding last visit and created columns such as “created.

There were a total of 137 observations that had no information regarding total lifetime value, days since last pay, product/subscription type nor subscription duration days. (see missingno matrix below for the 137 observations)

Dropping the 137 observations got rid of all the observations with non-realistic starting time



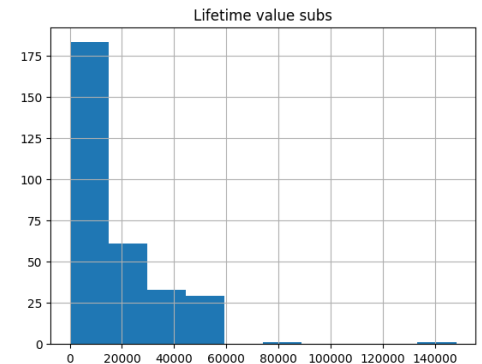
On business metrics

A short presentation of metrics to the owner was made. He liked some of the findings such as the 85-day early churn window (“We knew that we were bad at catching trial members but this is real bad”) and subscription duration patterns.

The owner will contact the payment processor handling non-POS transactions (swish). This in order to get a complete revenue picture by combining subscription payments (LTV), point-of-sale purchases, and external payment processor data.

It is unclear whether the bank will cooperate with this endeavour.

Additionally signups (based on df1 “created”) and cancelations (based on df6 cancelation y m) were combined into a time series graph showing net member growth.



Anonymization

Finally. Anonymization code was made using the uuid4 system in order to replace name with encoded strings. See attached .csv for an example of the final merged anonymized data.

As there is more cleaning to be done, such as removing staff and legacy members, this anonymized file should only be seen as a demonstration of current progress.

